# Project Documentation Report

## Smart library request workflow

| | |
|---|---|
| **Date** | 6 April 2026 |
| **Team ID** | SWTID-2026-9601 |
| **Project Name** | Smart library request workflow |
| **Maximum Marks** | 2 Marks |

## Team Members

| S.No | Name | RegNo |
|---|---|---|
| 1 | MUHMED JASSIR S | 812823205049 |
| 2 | MANIKANDAN S | 812823205046 |
| 3 | KISHOR L | 812823205039 |
| 4 | NITHISH M | 812823205053 |

# 1. INTRODUCTION:

## Project Purpose:

The purpose of the Smart Library Request Workflow is to automate and simplify the process of requesting, approving, and managing library resources using ServiceNow. It aims to reduce manual effort, eliminate errors, speed up request handling, and ensure efficient tracking and availability of library materials while improving user experience and productivity.

## Project Overview:

The Smart Library Request Workflow is an automated solution developed using ServiceNow to streamline the process of requesting, approving, and managing library resources. The system replaces traditional manual methods with a centralized digital platform, allowing users to easily submit requests for books or other materials through a self-service portal.

Once a request is submitted, the system automatically validates the data, checks resource availability, and routes the request for approval if required. Upon approval, tasks are generated for issuing or procuring the requested items. Throughout the process, users receive real-time notifications about the status of their requests.

## Key Objectives & Expected Outcomes:

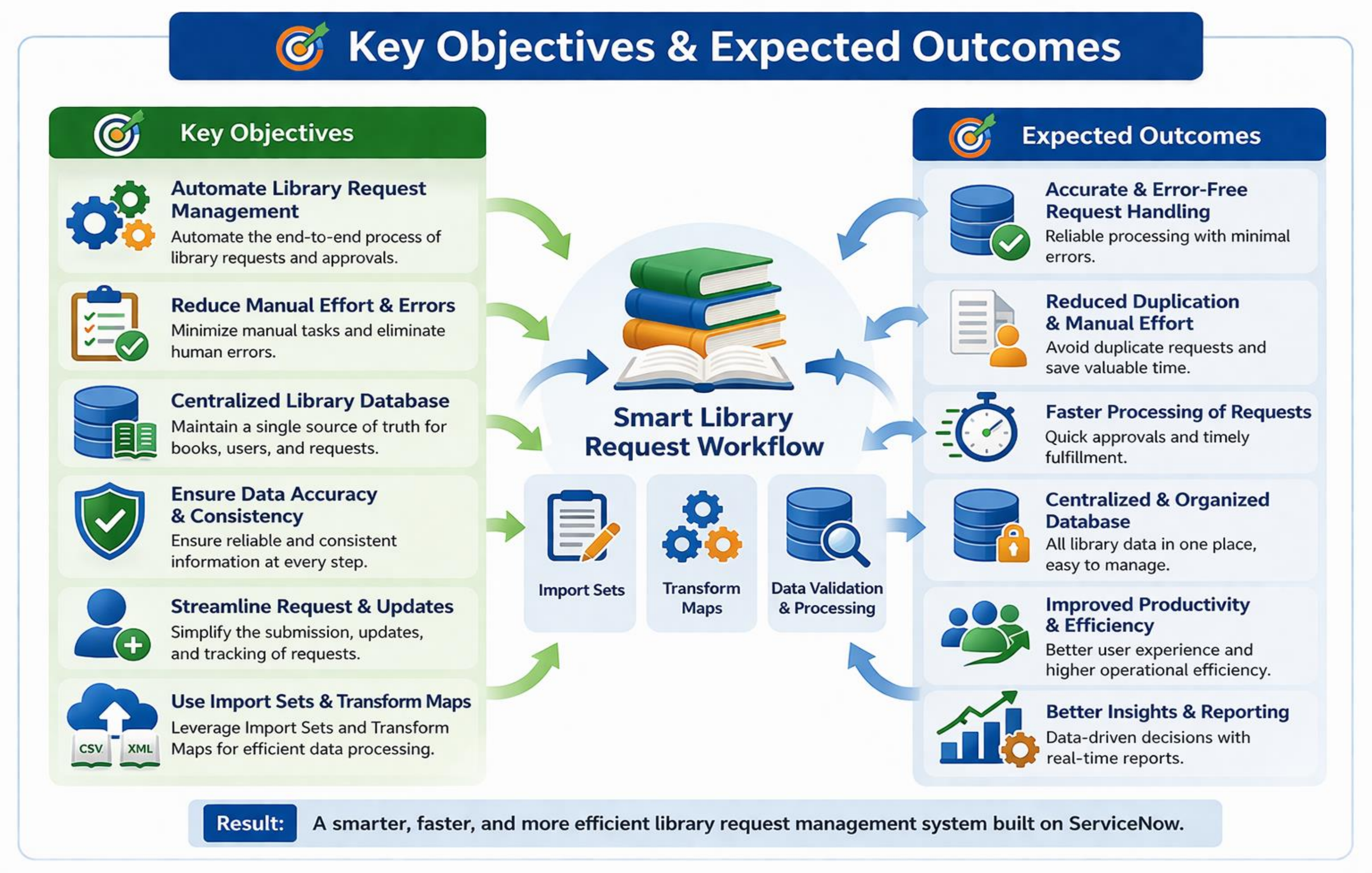
### Key Objectives:

The key objectives of this project are to automate the management of library requests using ServiceNow and reduce dependency on manual processes. The project focuses on creating a centralized system for handling book and resource requests while ensuring data accuracy, consistency, and proper tracking. It also aims to streamline request submission, validation, approval, and fulfillment by using automated workflows, thereby improving efficiency and user experience.

---

### Expected Outcomes:

The expected outcomes of this project include accurate and error-free handling of library requests with reduced duplication and minimal manual effort. The system enables faster processing of requests and provides a centralized, well-organized database for managing library resources. It also improves productivity and operational efficiency by ensuring easy access to up-to-date and reliable information on books, requests, and availability using ServiceNow.



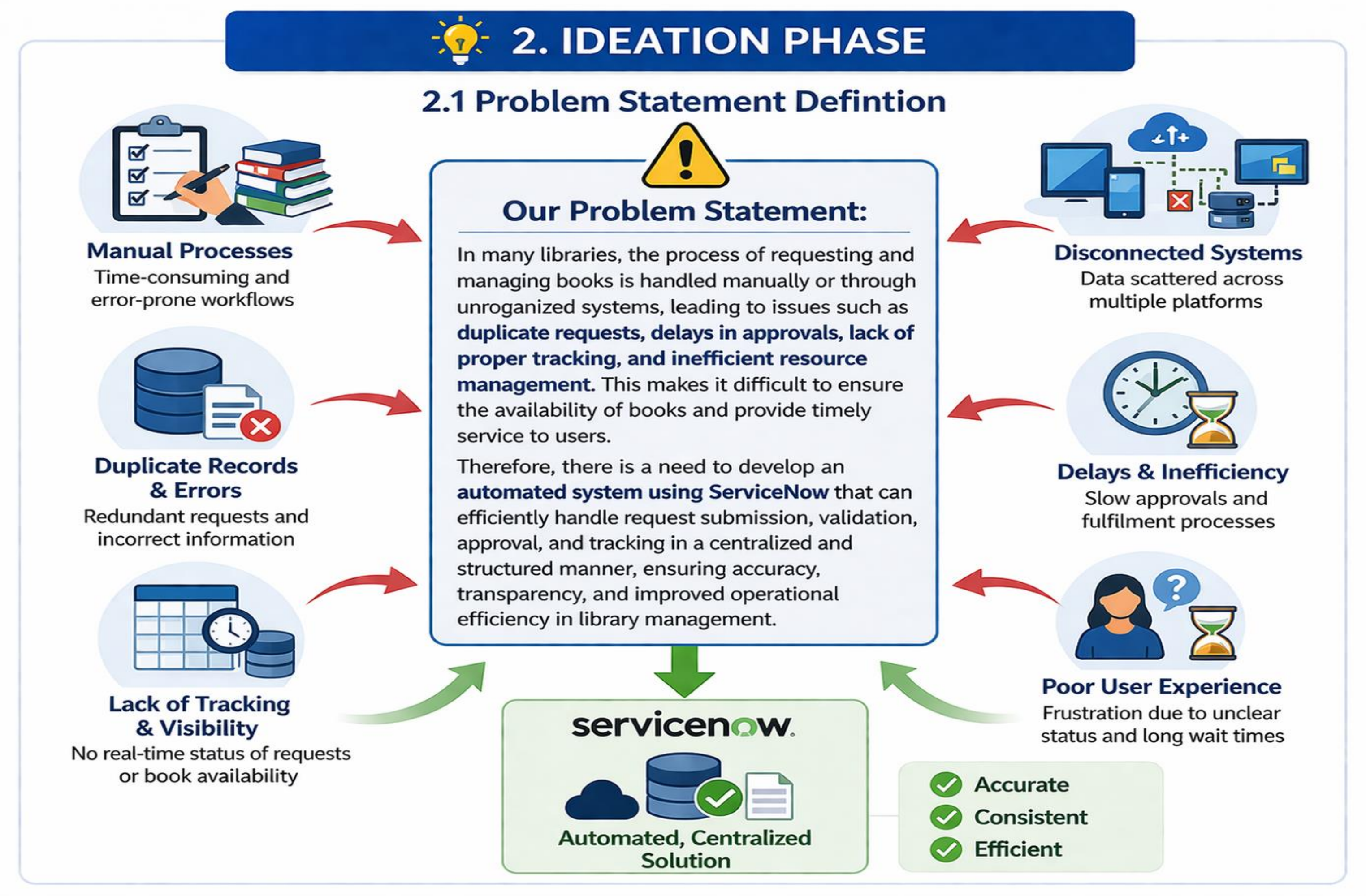
# 2. IDEATION PHASE:

## Problem Statement Definition

### Our Problem Statement:

In many libraries, the process of requesting and managing books is handled manually or through unorganized systems, leading to issues such as duplicate requests, delays in approvals, lack of proper tracking, and inefficient resource

management. This makes it difficult to ensure the availability of books and provide timely service to users.



## Problem Statements Identified

### PS-1

**Problem Statement (PS): PS-1**
**I am (Customer):** A librarian or library system administrator
**I'm trying to:** manage and process book/resource requests submitted by users through the system
**But:** the current process is manual, time-consuming, and lacks proper tracking, leading to duplicate requests and delays in approvals
**Because:** there is no automated workflow to validate, approve, and manage requests efficiently
**Which makes me feel:** frustrated and inefficient due to repetitive manual work and lack of system transparency

---

### PS-2

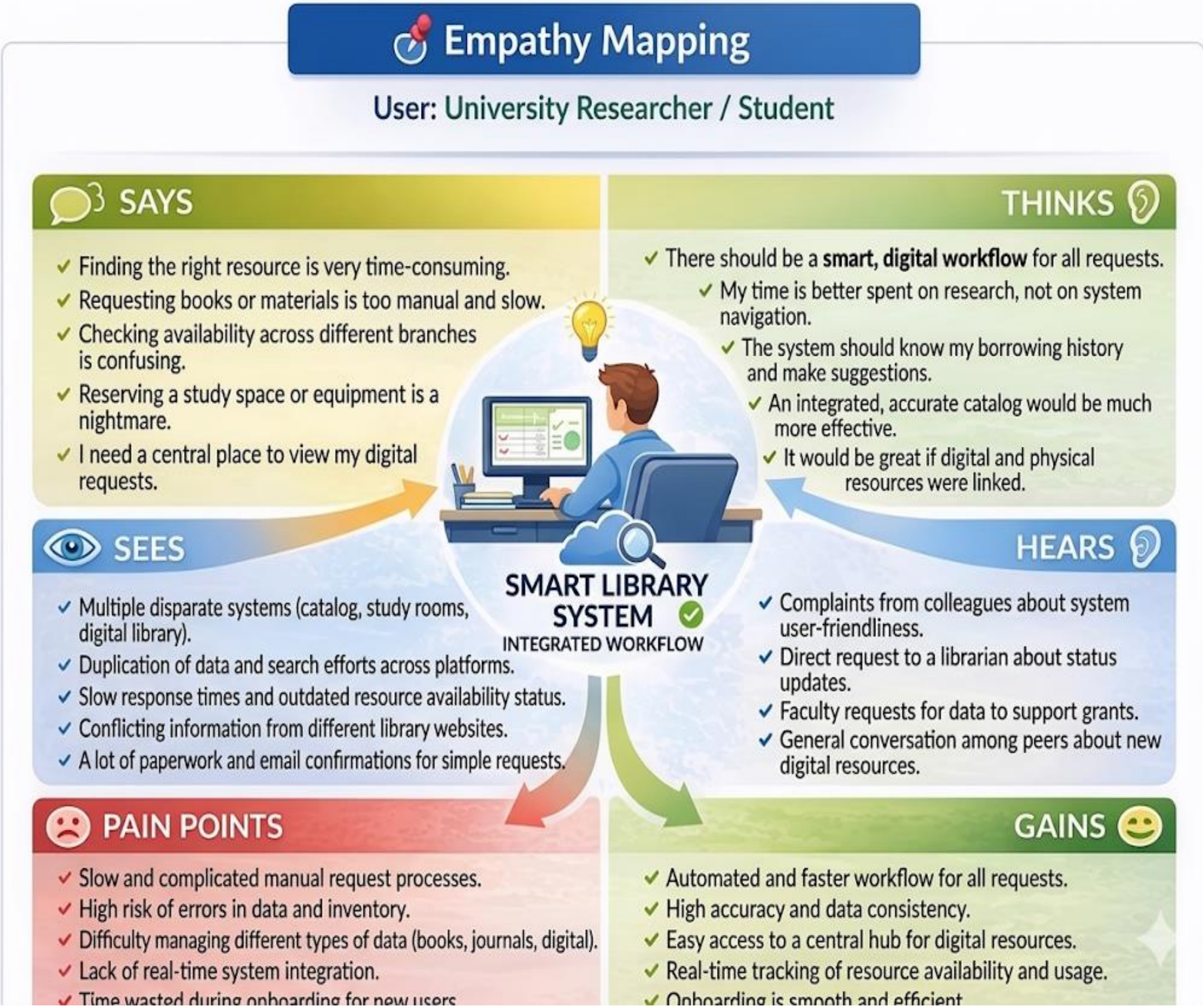
**I am (Customer):** A librarian or library system administrator
**I'm trying to:** manage and process book/resource requests submitted by users
**But:** the current process is manual, time-consuming, and lacks proper tracking, leading to duplicate requests and delays in approvals
**Because:** there is no automated workflow to efficiently validate, approve, and manage requests
**Which makes me feel:** frustrated and inefficient due to repetitive manual tasks and lack of system transparency

---

### PS-3

**I am (Customer):** A library management team member / organization user
**I'm trying to:** maintain a consistent and reliable database of library resources and requests
**But:** the system contains duplicate requests, outdated records, and inconsistent data
**Because:** there is no proper validation and automated workflow for managing library data
**Which makes me feel:** concerned about data accuracy, tracking, and effective decision-making

**Empathy Map:**


Empathy Mapping
User: University Researcher / Student
SAYS
Finding the right resource is very time-consuming.
Requesting books or materials is too manual and slow.
Checking availability across different branches is confusing.
Reserving a study space or equipment is a nightmare.
I need a central place to view my digital requests.
THINKS
There should be a smart, digital workflow for all requests.
My time is better spent on research, not on system navigation.
The system should know my borrowing history and make suggestions.
An integrated, accurate catalog would be much more effective.
It would be great if digital and physical resources were linked.
SMART LIBRARY SYSTEM
INTEGRATED WORKFLOW
SEES
Multiple disparate systems (catalog, study rooms, digital library).
Duplication of data and search efforts across platforms.
Slow response times and outdated resource availability status.
Conflicting information from different library websites.
A lot of paperwork and email confirmations for simple requests.
HEARS
Complaints from colleagues about system user-friendliness.
Direct request to a librarian about status updates.
Faculty requests for data to support grants.
General conversation among peers about new digital resources.
PAIN POINTS
Slow and complicated manual request processes.
High risk of errors in data and inventory.
Difficulty managing different types of data (books, journals, digital).
Lack of real-time system integration.
Time wasted during onboarding for new users.
GAINS
Automated and faster workflow for all requests.
High accuracy and data consistency.
Easy access to a central hub for digital resources.
Real-time tracking of resource availability and usage.
Onboarding is smooth and efficient.

## Brainstorming and Idea Prioritization:

### 1. Problem Statement

The project focuses on automating employee data import into ServiceNow to reduce manual work, minimize errors, and maintain accurate records. It also includes designing a smart library request workflow to streamline request handling and improve efficiency.

---

### 2. Brainstorming

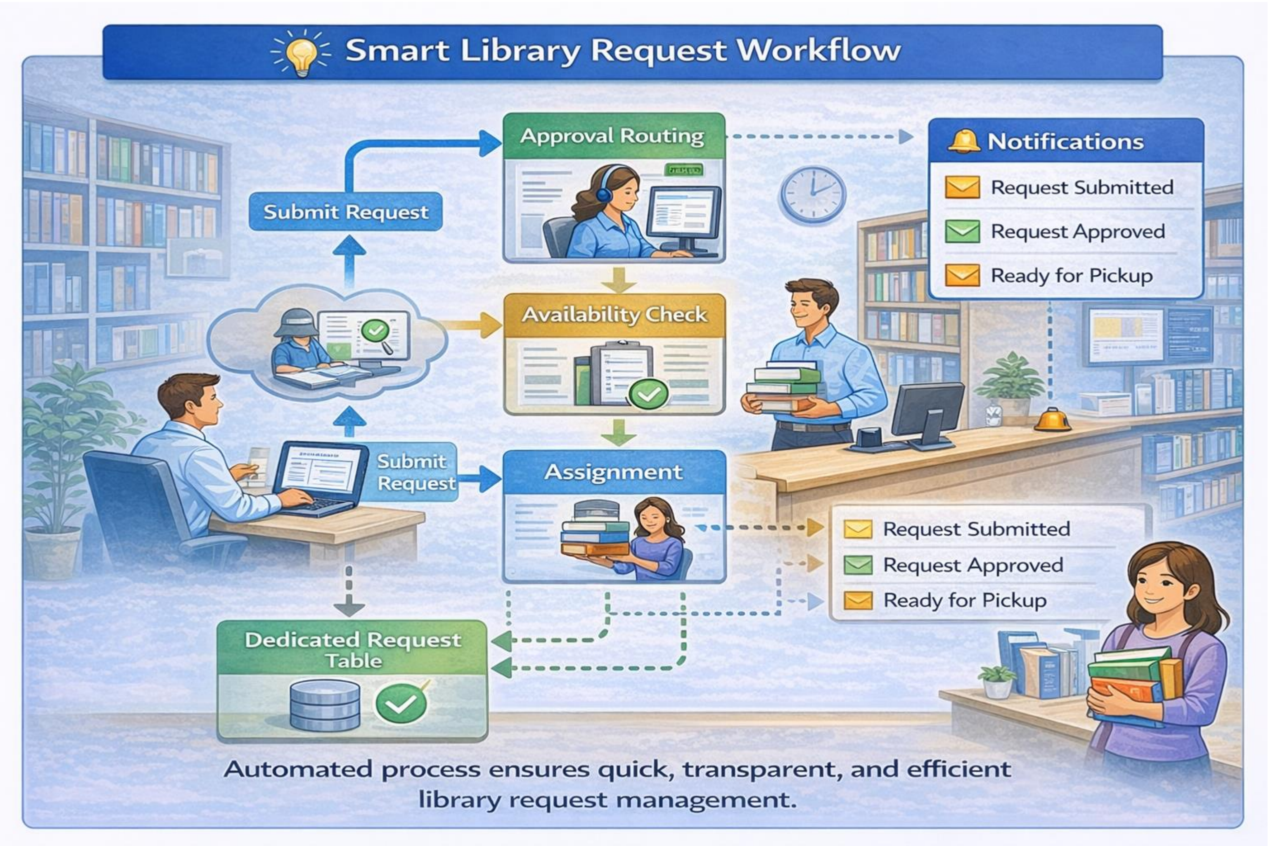
- For the smart library request workflow, requests can be submitted through forms or APIs, processed using Flow Designer or Business Rules, and stored in a custom or existing table.
- For employee data automation, import methods include Scheduled Import Sets, REST API, or Manual Upload, with processing via Transform Maps or Business Rules, and storage in the sys_user or a Custom Table.

---

### 3. Prioritization

The selected approach is **Import Sets + Transform Maps + sys_user Table**. This combination is chosen for its better performance, higher accuracy in data handling, and ease of implementation within the system.

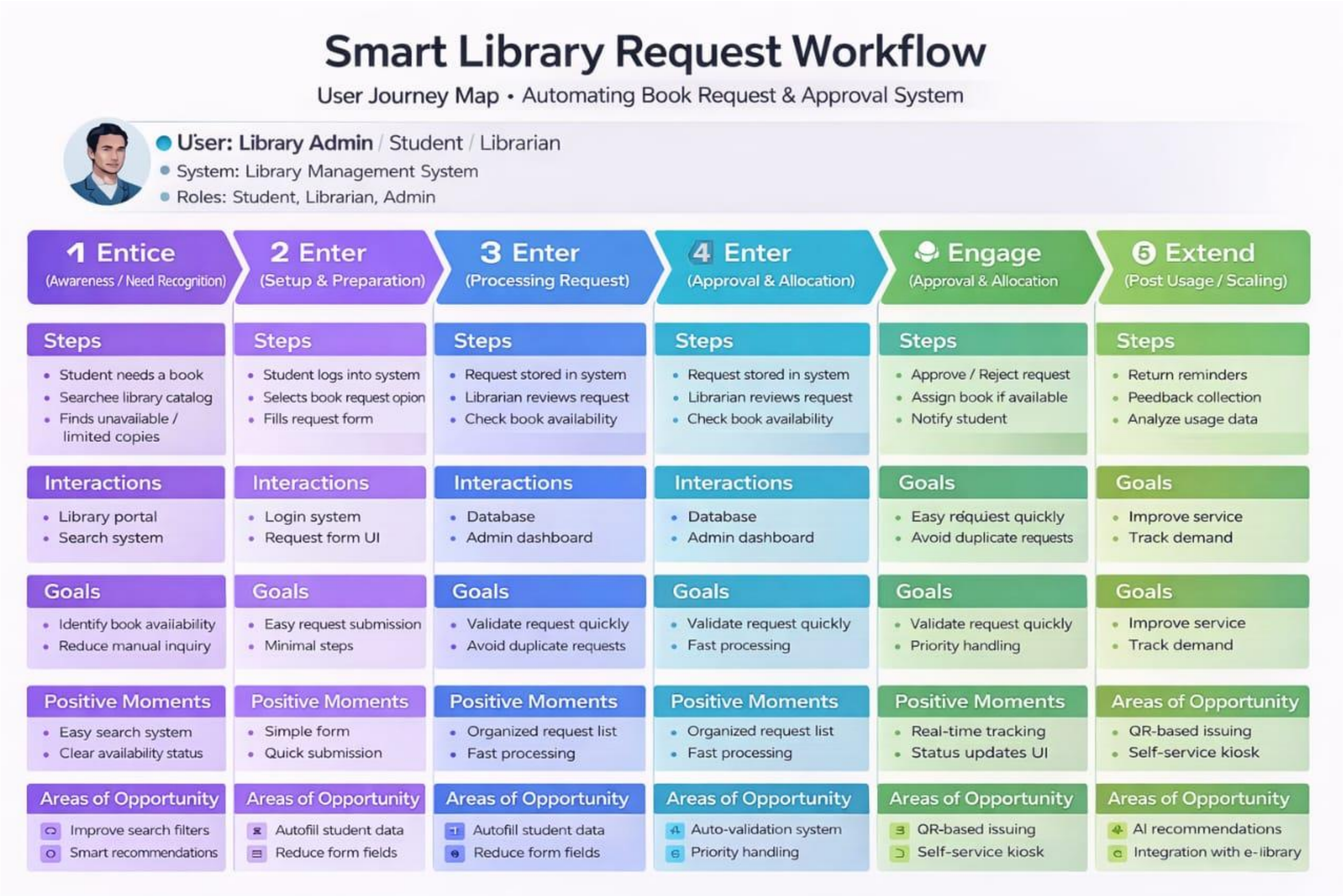
### Step-1: Team Gathering, Collaboration and Select the Problem statement.



Step-2: Brainstorm, Idea Listing and GroupiStep-3: Idea Prioritization

# 3. REQUIREMENT ANALYSIS:

## 3.1 Customer Journey Map



## Solution Requirements:

### Functional Requirements:

Following are the functional requirements of the proposed solution.

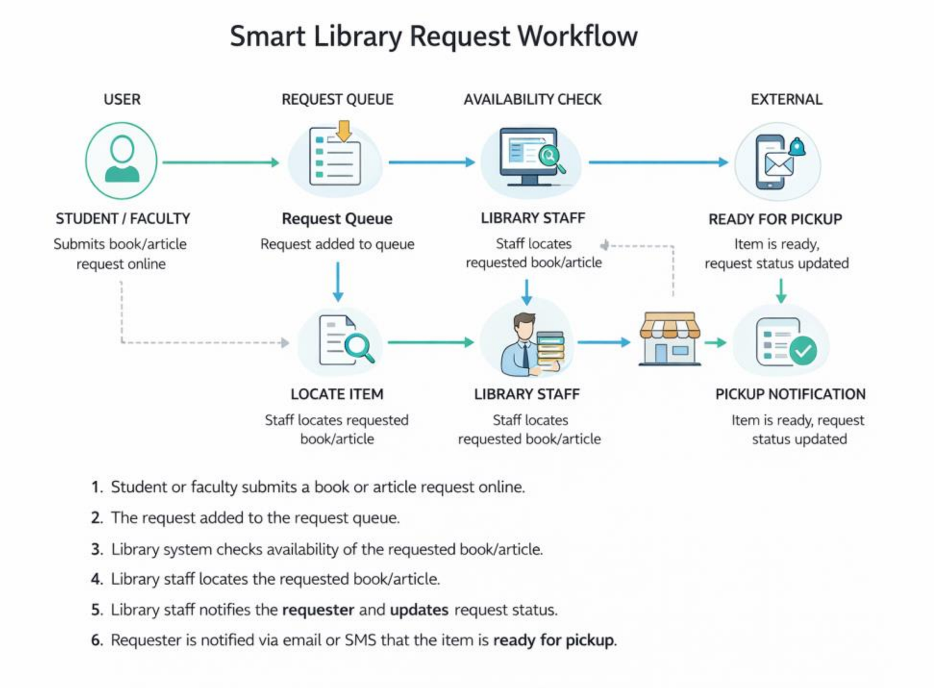
| FR No. | Functional Requirement | Sub Requirement |
|---|---|---|
| FR-1 | Book Request Import | Upload book request data through CSV/XLSX file<br><br>Automatic creation of import table (import_library_request) |
| FR-2 | Data Transformation & Mapping | Create Transform Map between import_library_request and library_request table<br><br>Map basic fields (Book Title, Author, Category) |
| FR-3 | Data Validation & Processing | Execute Transform Map to insert records<br><br>Validate correct data insertion into library_request table |
| FR-4 | Incident Form Automation | Display user details on Library Request form<br><br>Show Department using dot-walking. |
| FR-5 | Dynamic Data Display | Auto-update fields when "Requested By" user changes |
| FR-6 | Security & Access Control | Restrict request import access to authorized usersRestrict approval access to Librarian role.Ensure data visibility based on roles (ACLs) |

## Non-functional Requirements:

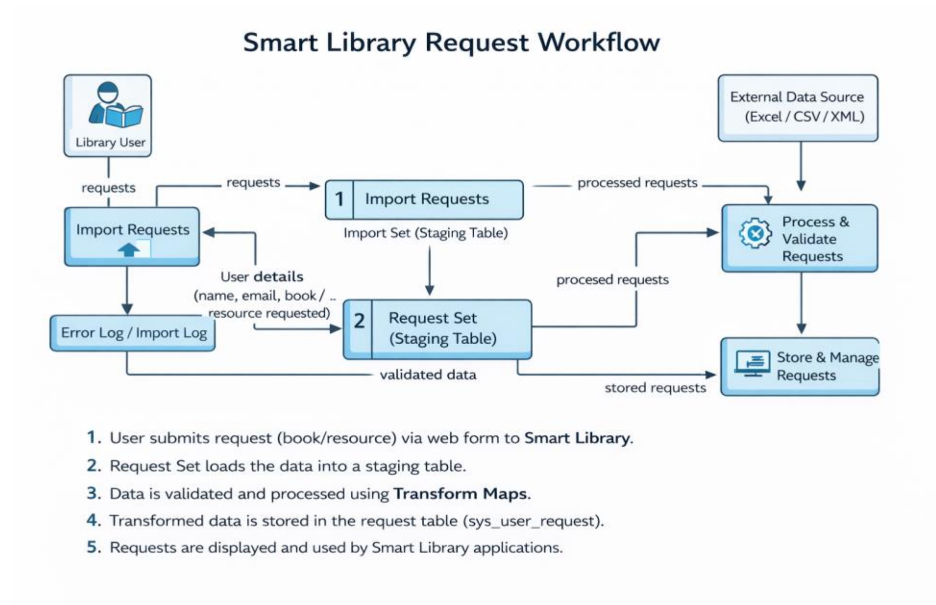
| FR No. | Non-Functional Requirement | Description |
|---|---|---|
| NFR-1 | **Usability** | The system should provide a user-friendly interface for submitting and tracking library requests. Users should be able to request books easily, and the request status should be displayed clearly without additional navigation.. |
| NFR-2 | **Security** | Only authorized users (Librarian/Admin roles) should be allowed to approve or manage requests. Data visibility must follow Role-Based Access Control (RBAC) and Access Control Lists (ACLs). |
| NFR-3 | **Reliability** | The system should handle invalid or incomplete request data gracefully. Errors during request submission or processing should be logged without affecting other requests. |
| NFR-4 | **Performance** | The system should process library requests quickly without affecting platform performance. Request forms and status updates should load instantly without delay. |
| NFR-5 | **Availability** | The system should be available at all times with minimal downtime. Request submission, approval workflow, and tracking features should function consistently. |
| NFR-6 | **Scalability** | The system should support both small and large numbers of requests (e.g., hundreds or thousands of requests) without performance degradation and scale efficiently as usage increases. |

### 3.3 Data Flow Diagrams:

Level 1 DFD




Level 2 DFD



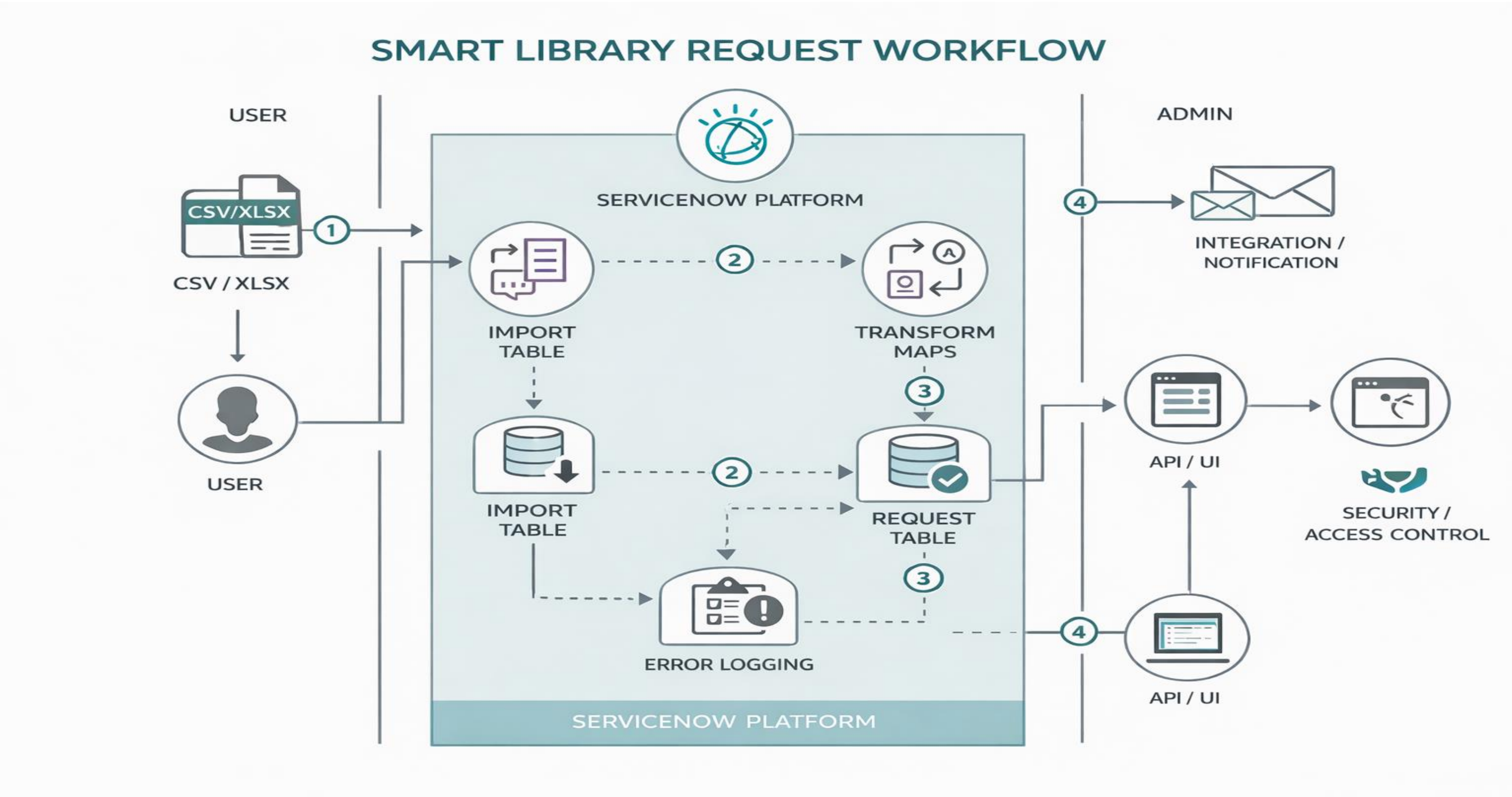
**User Stories:**

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria |
|---|---|---|---|---|
| Service Now Admin | Data Import Setup | USN-1 | As a Service Now Admin, I want to upload employee data (CSV/XLSX) so that it can be stored in the system | File is uploaded successfully and data appears in import table |
| ServiceNow Admin | Request Import Setup | USN-2 | As an Admin, I want the system to automatically create an import table (`import_library_request`) | import table is created with all uploaded records |
| ServiceNow Admin | Request Transformation | USN-3 | Create a Transform Map to move data to request table | Transform map is created and linked to correct tables |
| ServiceNow Admin | Request Transformation | USN-4 | Map fields (Requester Name, Email, Book Title, Author, Request Type, Department) | All fields are correctly mapped without errors |
| ServiceNow Admin | Request Validation | USN-5 | Validate transformed data for accuracy | Data is correctly inserted into system tables (e.g., user/request tables) |
| ServiceNow Admin | Error Handling | USN-6 | Log errors for missing or incorrect data | Errors are logged without stopping the process |
| ServiceNow Admin | Form Configuration | USN-7 | Configure forms to display employee and request details | Fields are added and visible on form |

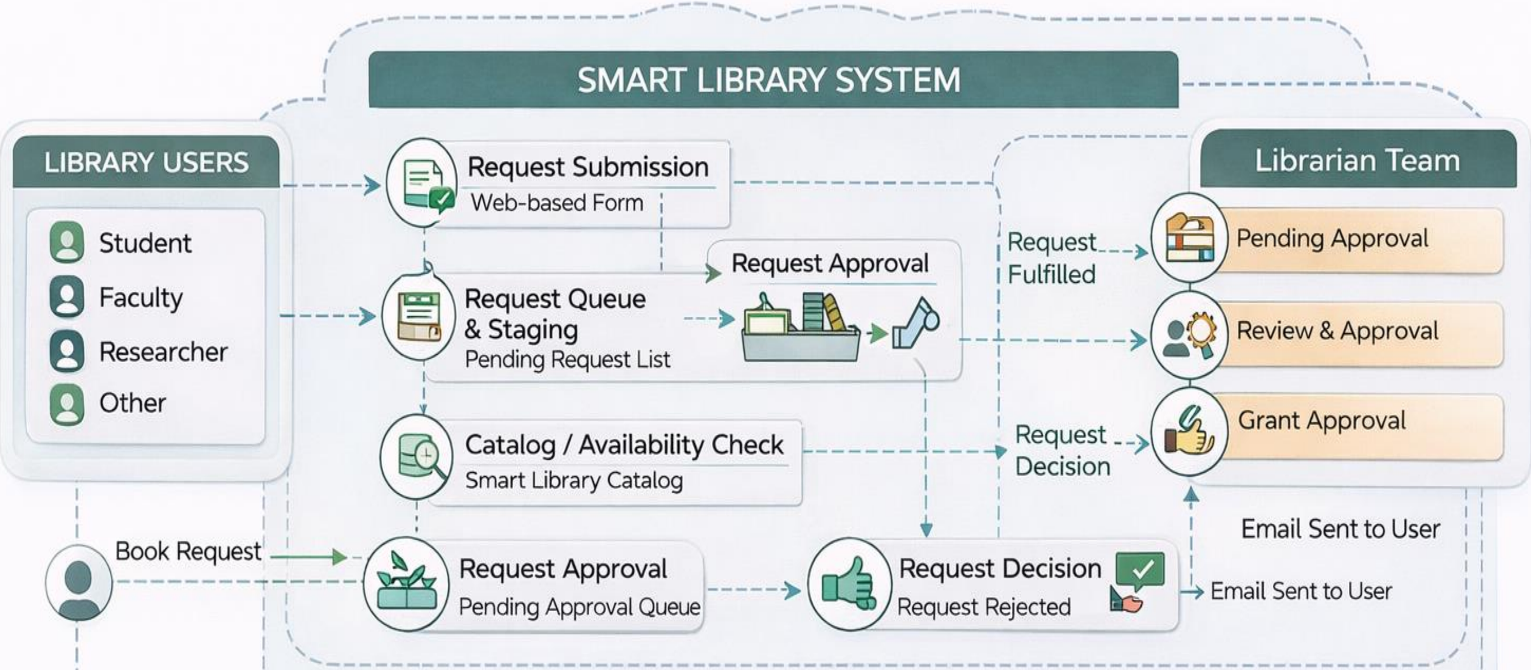
## Technology Stack:

### Technical Architecture:

The **Smart Library Request Workflow** uses ServiceNow to automate data import, transformation, validation, and request management through forms, Import Sets, and Transform Maps.



# Smart Library Request Workflow –
## Automating Book Requests & Approval Process




| # | Component | Description | Technology |
|---|---|---|---|
| 1 | Request Submission | User Request Form | Web-based Form |
| 2 | Request Queue & Staging | Pending Request List | Request Queue Table |
| 3 | Catalog / Availability Check | Smart Library Catalog | Digital Catalog Database |
| 4 | Request Approval | Pending Approval Queue | Approval Workflow |
| 5 | Request Decision | Approve or Reject Request | Request Management UI |
| 6 | Automatic Fulfillment | Automatic Book Handling | Fulfillment Automation |

**Table-1 : Components & Technologies:**

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1 | User Interface | User-facing portal for students, faculty, and researchers to request books, track request status, and view availability. Admin dashboard for librarians to manage requests, approvals, and inventory. | HTML5, CSS3, JavaScript, React JS, Bootstrap |
| 2 | Application Logic-1 | Core business logic for handling book requests, approvals, request prioritization, and request validation. | Python, Flask / Java Spring Boot |
| 3 | Application Logic-2 | Voice-based request submission allowing users to search and request books using speech commands. | IBM Watson Speech to Text |
| 4 | Application Logic-3 | AI assistant to answer user queries related to book availability, due dates, fines, and request status. | IBM Watson Assistant |
| 5 | Database | Stores book details, user information, request history, approval status, and borrowing records. | MySQL / PostgreSQL |
| 6 | Cloud Database | Centralized cloud storage for library data access across departments and backup management. | IBM Db2 on Cloud / IBM Cloudant |
| 7 | File Storage | Stores digital resources such as e-books, research papers, user documents, and library reports. | IBM Cloud Object Storage / Local File System |
| 8 | External API-1 | Sends notifications to users for request approval, availability updates, due reminders, and return alerts. | Twilio API, SendGrid API |
| 9 | External API-2 | Integration with external library catalogs or academic databases for checking book availability. | Google Books API / Open Library API |
| 10 | Machine Learning Model | Predicts book demand, recommends books to users, and analyzes borrowing patterns. | IBM Watson Machine Learning, Scikit-learn |
| 11 | Infrastructure (Server / Cloud) | Cloud deployment for hosting library system, database, AI modules, and request workflow services. | IBM Cloud, Docker, Kubernetes, Cloud Foundry |

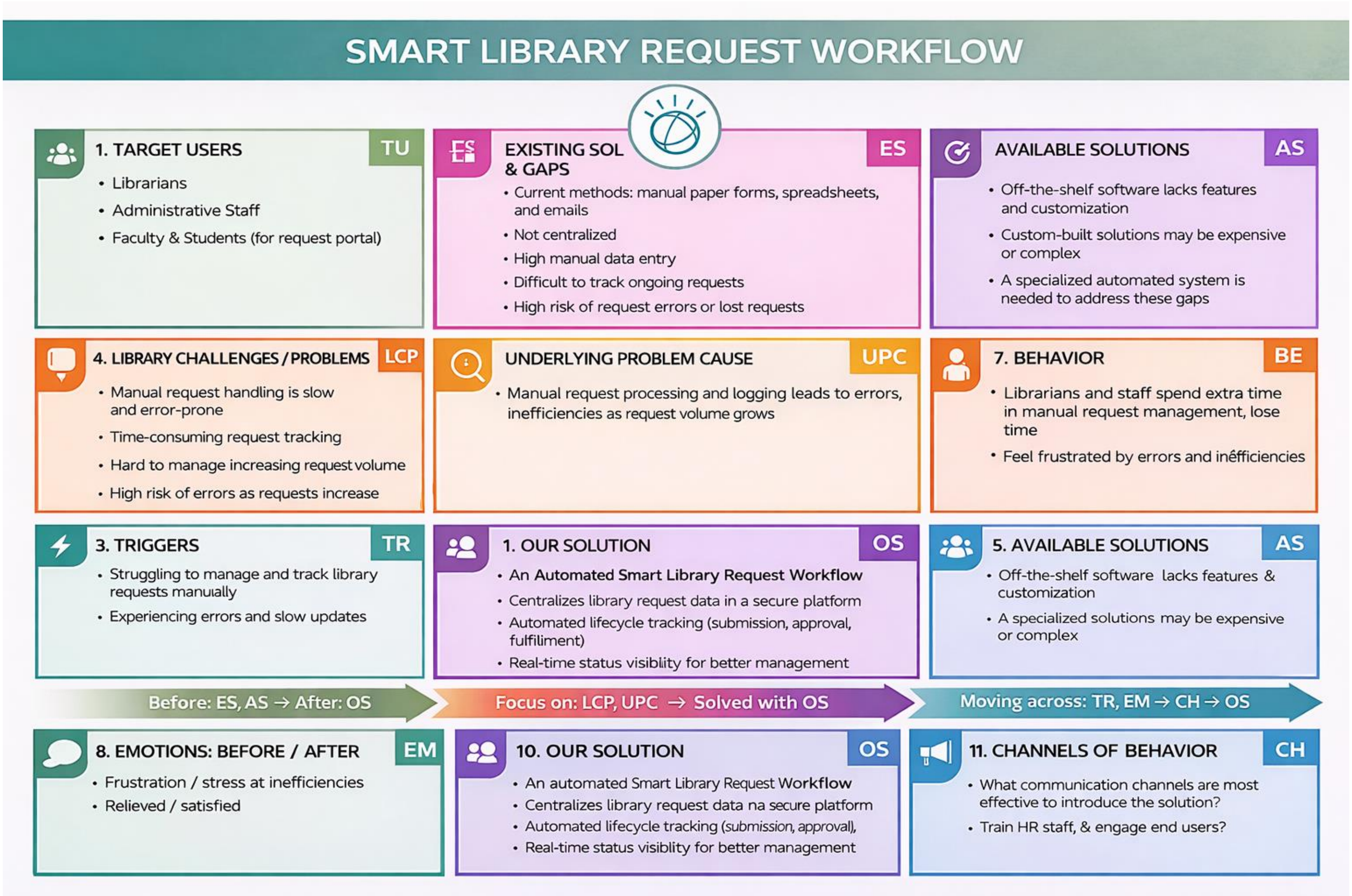
**Table-2: Application Characteristics:**

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1 | Open-Source Frameworks | Open-source frameworks are used for developing the frontend, backend, APIs, and AI-based recommendation modules of the smart library request workflow system. | React JS, Flask, Spring Boot, Node.js, Scikit-learn |
| 2 | Security Implementations | Secure login and role-based access for students, faculty, and librarians. User credentials and request data are protected during transmission and storage. | SHA-256, AES Encryption, HTTPS, JWT Authentication, IBM IAM, OWASP Security Standards |
| 3 | Scalable Architecture | The system follows a 3-tier architecture including presentation, application, and database layers, allowing independent scaling for request handling and catalog management. | Microservices Architecture, Docker, Kubernetes |
| 4 | Availability | The smart library system ensures continuous availability using cloud infrastructure, backup databases, and distributed servers for handling multiple user requests. | Load Balancer, IBM Cloud Availability Zones, Distributed Servers |
| 5 | Performance | Fast response time for book search, request processing, and catalog browsing using caching and optimized queries. | Redis Cache, CDN, Optimized SQL Queries |
| 6 | AI-Based Recommendation | The system recommends books to users based on borrowing history, search patterns, and popular demand. | IBM Watson Machine Learning, Scikit-learn, Recommendation Algorithms |

# 4. PROJECTDESIGN

## Problem – Solution Fit:

Manual handling of library requests can lead to errors, delays, and difficulty in tracking records as the number of requests increases. The **Smart Library Request Workflow** automates the entire process by centralizing data, reducing manual effort, and ensuring accurate request handling. This improves efficiency, saves time, and provides better visibility and management of library requests within the system.



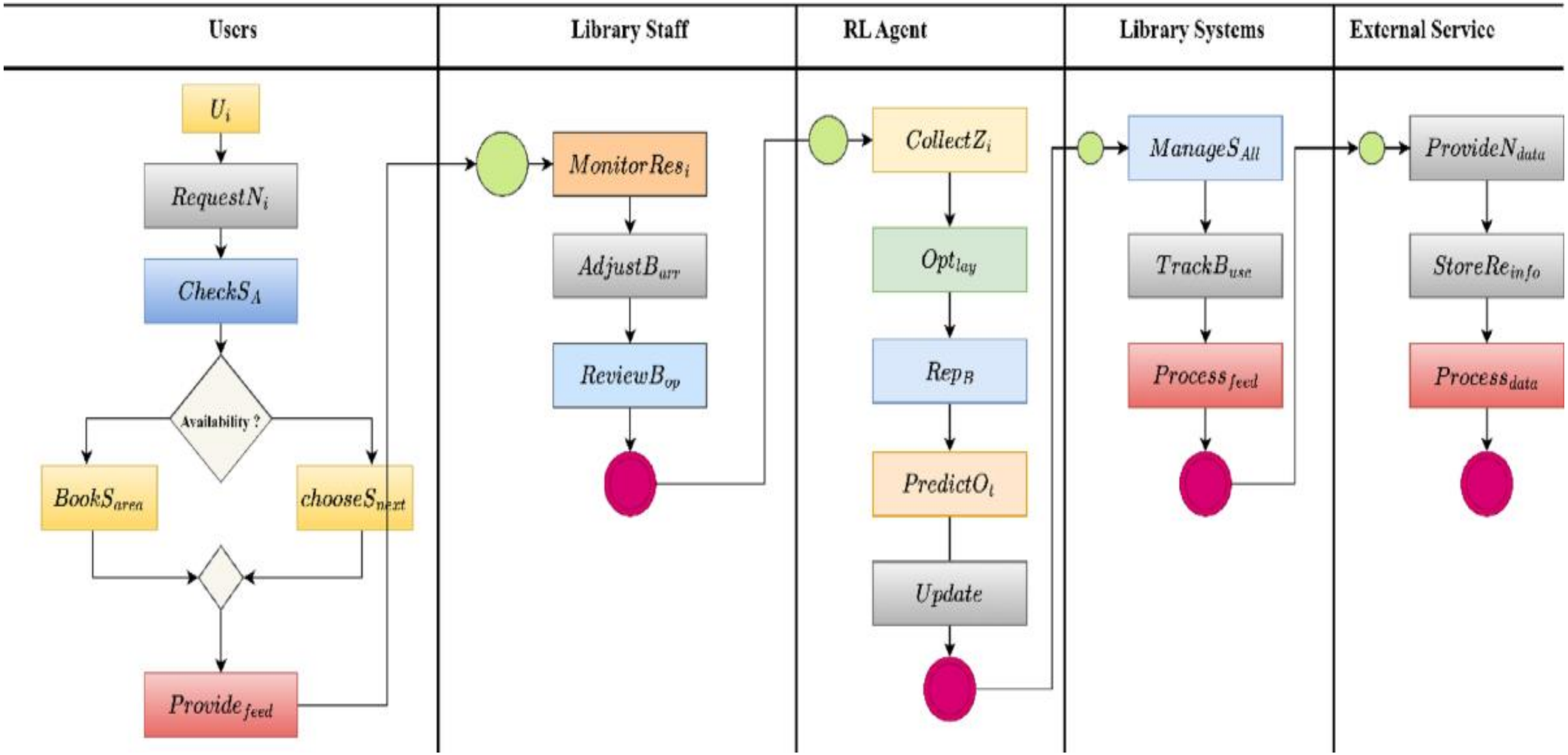
## Purpose:

- To develop an automated system for efficient library request management
- To reduce manual work and minimize human errors in handling requests
- To centralize all library request data in a single system
- To streamline processes like request submission, tracking, and approval
- To improve data accuracy, security, and accessibility
- To enhance overall efficiency and productivity in library operations

**Proposed Solution:**

| No. | Parameter | Description |
|---|---|---|
| 1. | Problem Statement (Problem to be solved) | Managing library requests manually is time-consuming and inefficient. Librarians face issues such as misplaced requests, delayed approvals, difficulty in tracking book availability, and lack of real-time updates for users. |
| 2. | Idea / Solution Description | The project proposes a Smart Library Request Workflow system where users can submit book requests digitally. The system automatically checks availability, routes approval to librarians, updates request status, and notifies users when books are available. |
| 3. | Novelty / Uniqueness | • Reduces manual paperwork using automation • Digital request tracking system Real-time availability and status updates • AI-based book recommendations • Automated notifications for approvals and returns |
| 4. | Social Impact / Customer Satisfaction | • Saves time for students and librarians • Improves accessibility to books and resources • Faster request processing • Enhances library user experience • Reduces manual errors and delays |
| 5. | Business Model (Revenue Model) | • Can be offered as Library Management Software (SaaS) • Subscription-based model for institutions • Paid customization for schools, colleges, and organizations • Premium features like AI recommendations and analytics |
| 6. | Scalability of the Solution | • Can handle large library databases • Easily expandable to include e-books and digital resources • Supports multiple libraries and departments • Suitable for schools, colleges, universities, and organizations |

### 4.3 Solution Architecture



## 5. PROJECT PLANNING & SCHEDULING

### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Before the system goes live, these planning steps ensure the "smart" infrastructure is ready to handle requests:

- **Goal Identification**: Define specific objectives like reducing manual check-outs or enhancing resource tracking.
- **Needs Assessment**: Evaluate current manual processes and feedback from staff and users.
- **Infrastructure Setup**: Deploy hardware such as **RFID tag readers**, cameras for facial recognition, and **IoT sensors**.
- **Database Configuration**: Establish master databases for **Books**, **Members**, and **Vendors**.
- **Workflow Automation Design**: Create digital triggers (e.g., via SmartSuite or Library Open Workflows) to handle approvals and notifications.

**Project Tracker, Velocity & Burndown Chart:**

| Sprint | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed | Sprint Release Date (Actual) |
|---|---|---|---|---|---|---|
| **Sprint-1** | 9 | 16 Days | 01 Mar 2026 | 15 Mar 2026 | - | - |
| **Sprint-2** | 7 | 16 Days | 16 Mar 2026 | 01 Apr 2026 | - | |

**Velocity:**

$$\boldsymbol{V} = \frac{\boldsymbol{SprintDuration}}{\;} \qquad \frac{\;}{\mathbf{8}} = \mathbf{2}$$

$$\therefore \boldsymbol{AV} = \frac{\boldsymbol{SprintDuration}}{\boldsymbol{Velocity}} \qquad \frac{\;}{\;}$$

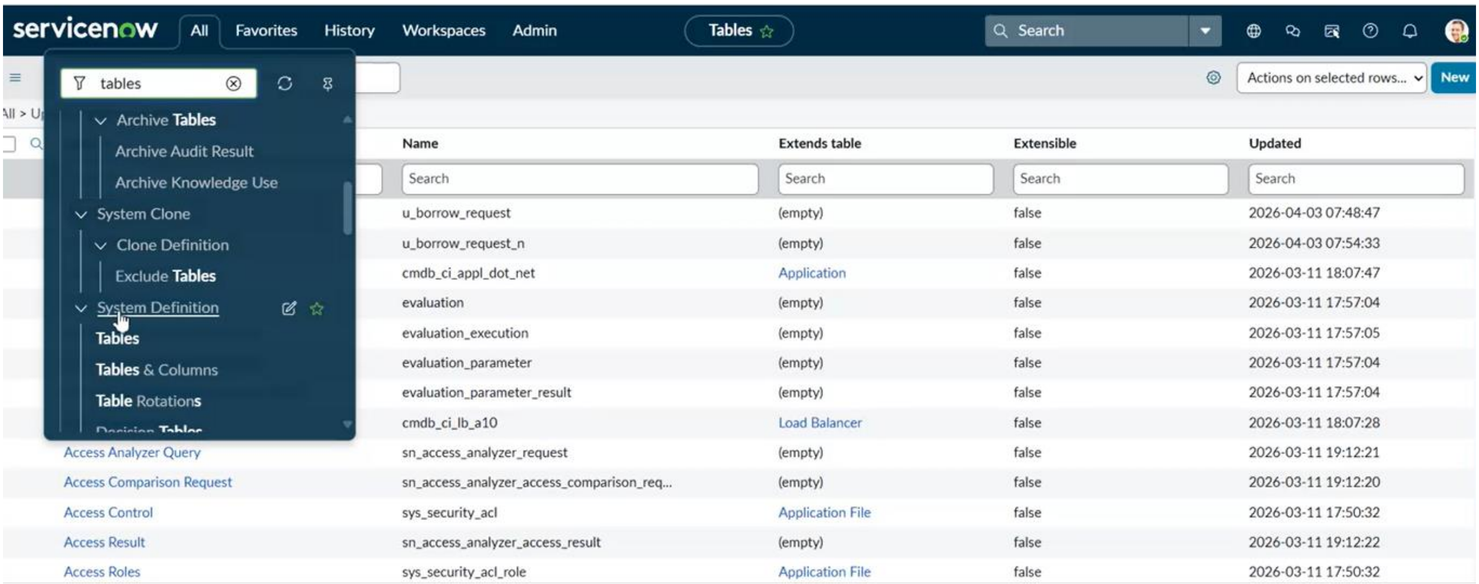
$$= \mathbf{16} = 2$$

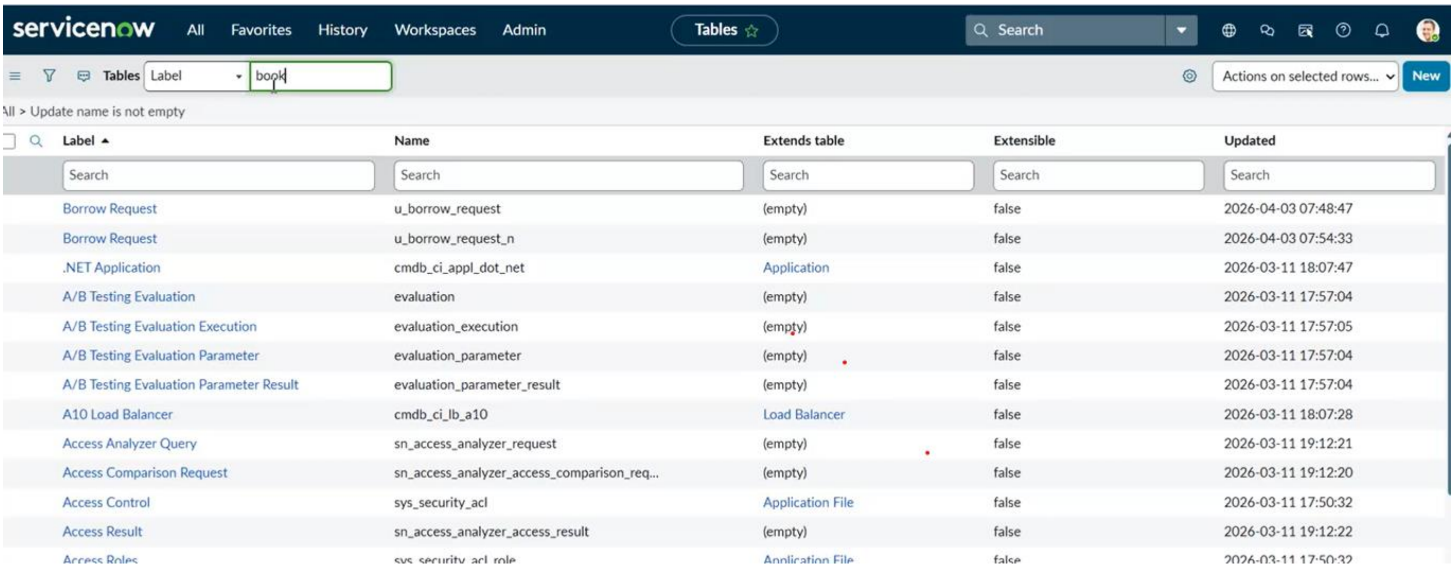
## 6. FUNCTIONALANDPERFORMANCETESTING:

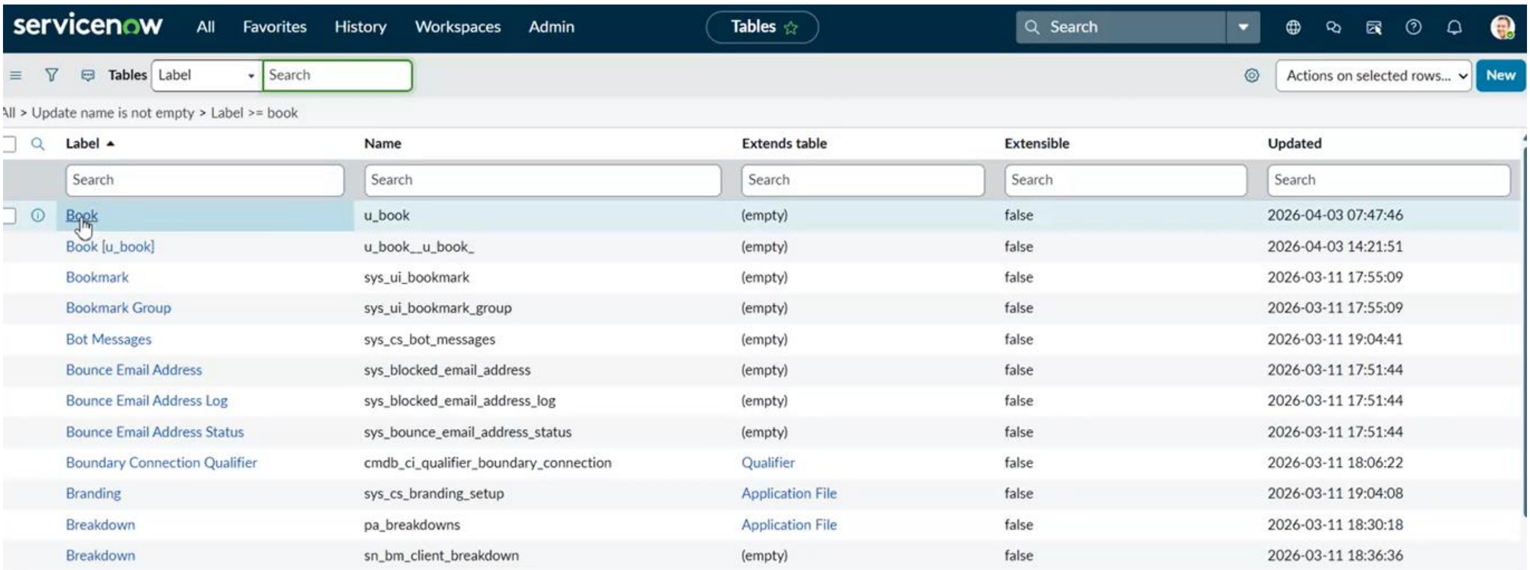
**Test Scenarios & Results**:

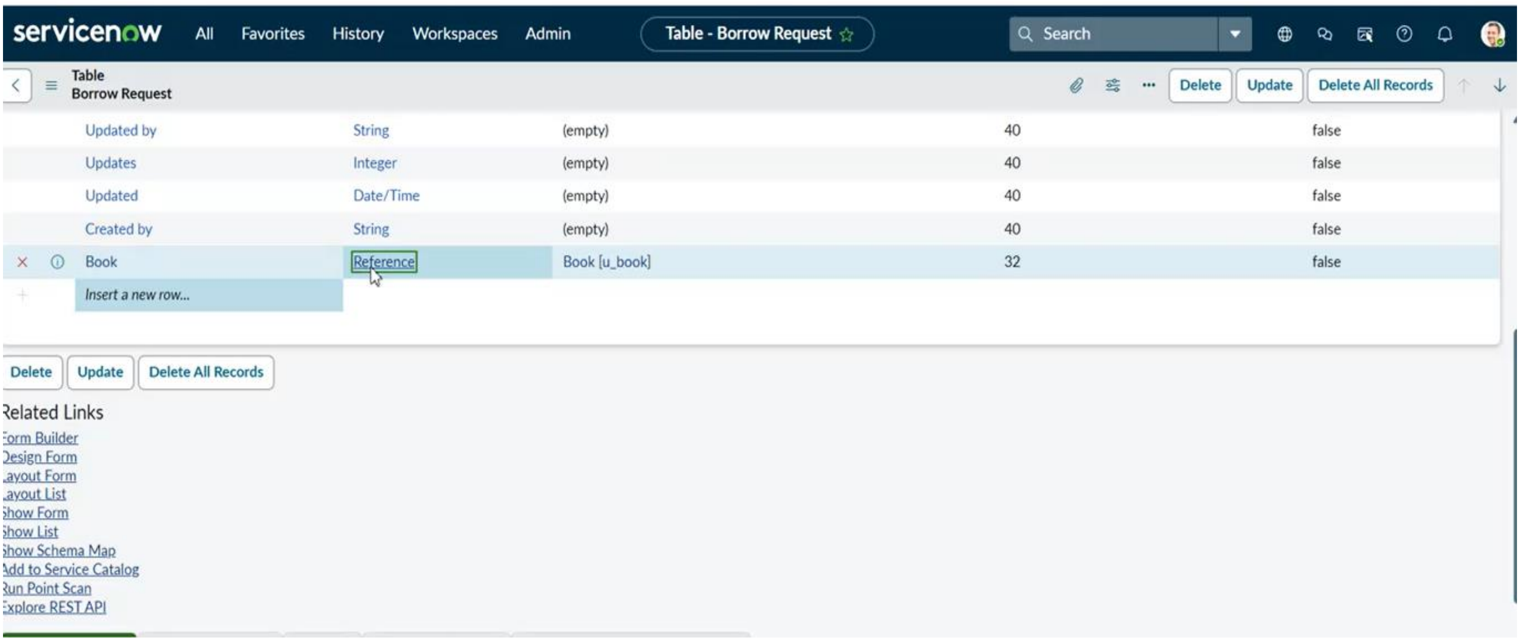
| Test Case ID | Scenario (What to test) | Test Steps (How to test) | Expected Result | Actual Result | Pass/Fail |
|---|---|---|---|---|---|
| FT-01 | Book Request Submission | 1. Login as user 2. Navigate to Request Book 3. Submit a request | Book request is successfully submitted and stored in the system | — | — |
| FT-02 | User Registration | 1. Open registration page 2. Enter valid details 3. Submit form | User account is created and confirmation is received | — | — |
| FT-03 | Login Functionality | 1. Enter valid username & password 2. Click login | User is logged in and redirected to dashboard | — | — |
| FT-04 | Request Approval Workflow | 1. Login as admin 2. View pending requests 3. Approve request | Request status changes to "Approved" | — | — |
| FT-05 | Request Tracking | 1. Login as user 2. Check request status | User can view real-time status (Pending/Approved/Rejected) | — | — |

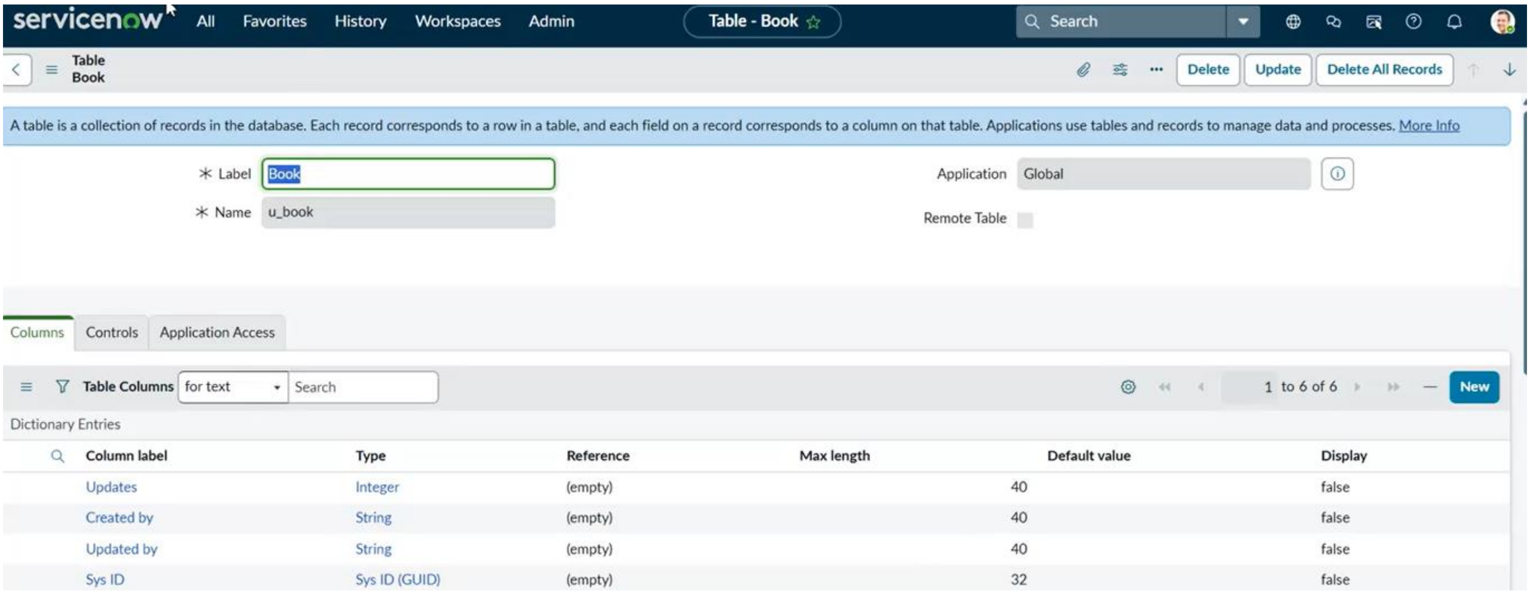
# RESULTS:

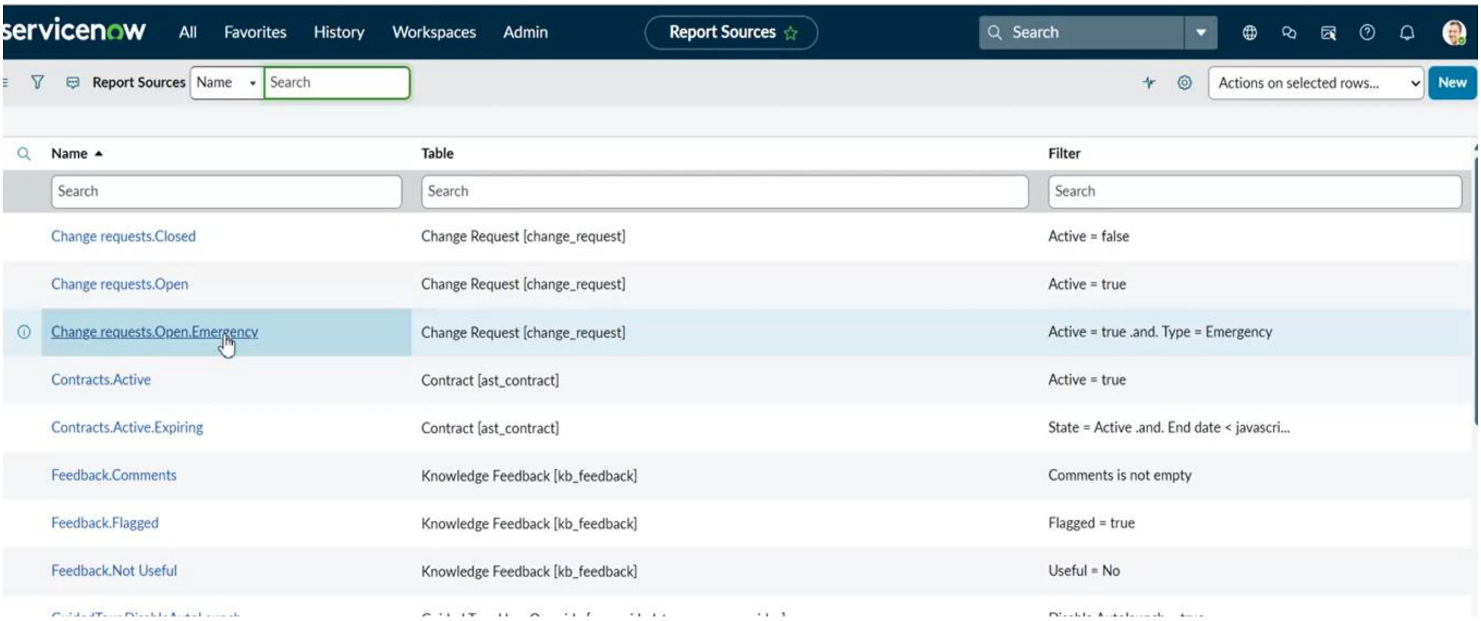


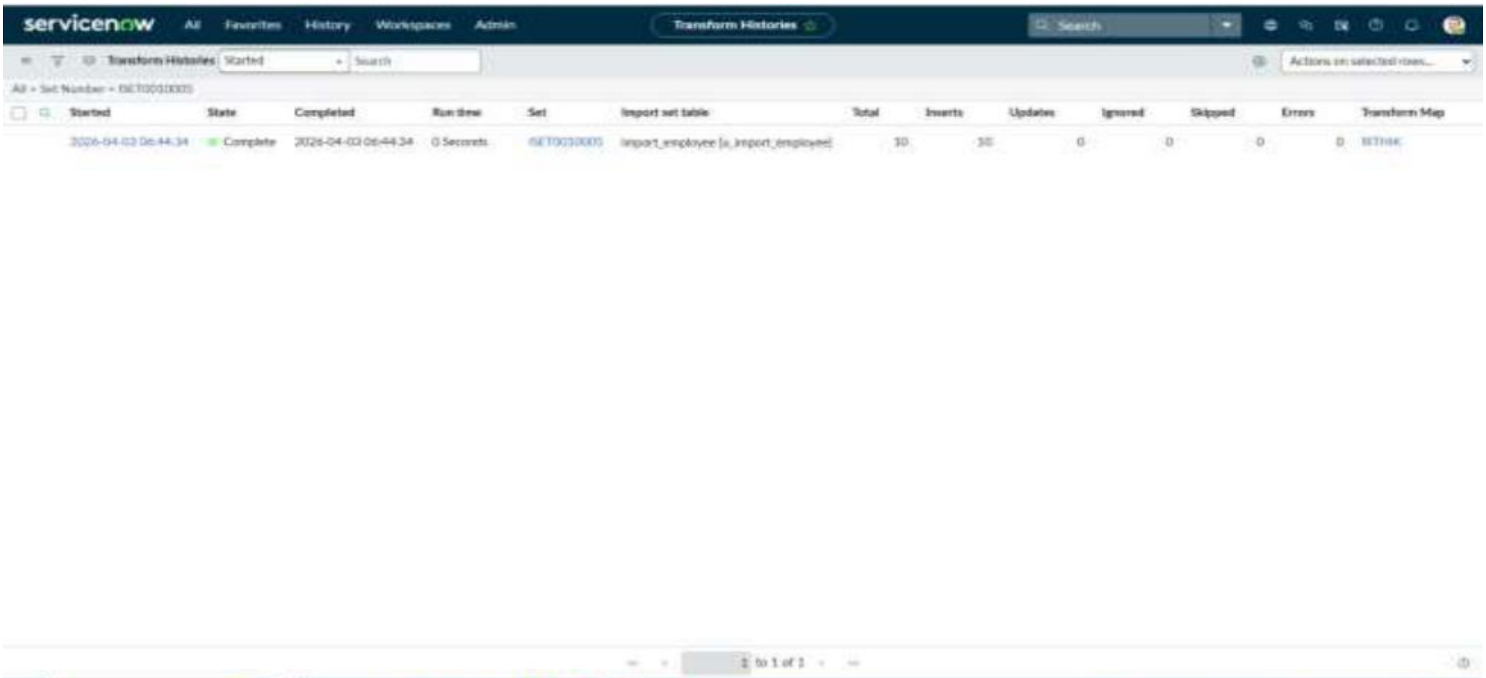

servicenow
All
Favorites
History
Workspaces
Admin
Tables
Search
Tables
Label
book
Actions on selected rows...
New
All > Update name is not empty
Label
Name
Extends table
Extensible
Updated
Borrow Request | u_borrow_request | (empty) | false | 2026-04-03 07:48:47
Borrow Request | u_borrow_request_n | (empty) | false | 2026-04-03 07:54:33
.NET Application | cmdb_ci_appl_dot_net | Application | false | 2026-03-11 18:07:47
A/B Testing Evaluation | evaluation | (empty) | false | 2026-03-11 17:57:04
A/B Testing Evaluation Execution | evaluation_execution | (empty) | false | 2026-03-11 17:57:05
A/B Testing Evaluation Parameter | evaluation_parameter | (empty) | false | 2026-03-11 17:57:04
A/B Testing Evaluation Parameter Result | evaluation_parameter_result | (empty) | false | 2026-03-11 17:57:04
A10 Load Balancer | cmdb_ci_lb_a10 | Load Balancer | false | 2026-03-11 18:07:28
Access Analyzer Query | sn_access_analyzer_request | (empty) | false | 2026-03-11 19:12:21
Access Comparison Request | sn_access_analyzer_access_comparison_req... | (empty) | false | 2026-03-11 19:12:20
Access Control | sys_security_acl | Application File | false | 2026-03-11 17:50:32
Access Result | sn_access_analyzer_access_result | (empty) | false | 2026-03-11 19:12:22

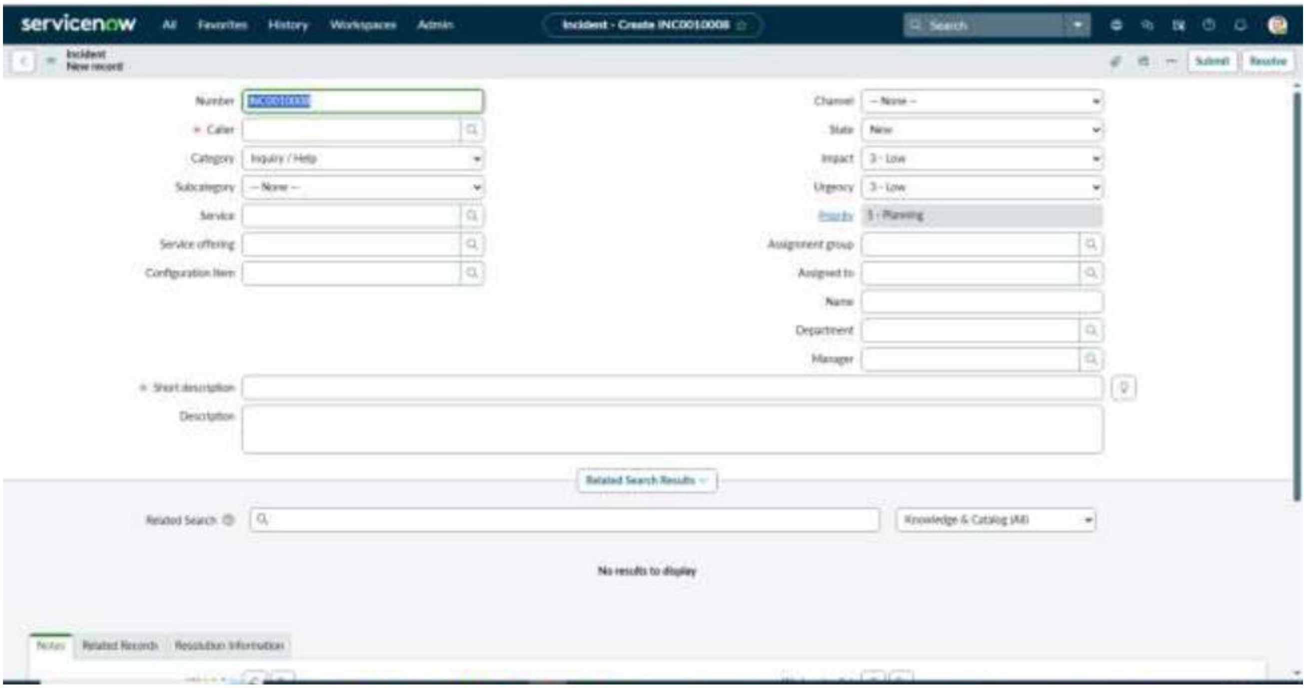

servicenow
All
Favorites
History
Workspaces
Admin
Tables
Search
Tables
Label
Search
Actions on selected rows...
New
All > Update name is not empty > Label >= book
Label
Name
Extends table
Extensible
Updated
Book | u_book | (empty) | false | 2026-04-03 07:47:46
Book [u_book] | u_book__u_book_ | (empty) | false | 2026-04-03 14:21:51
Bookmark | sys_ui_bookmark | (empty) | false | 2026-03-11 17:55:09
Bookmark Group | sys_ui_bookmark_group | (empty) | false | 2026-03-11 17:55:09
Bot Messages | sys_cs_bot_messages | (empty) | false | 2026-03-11 19:04:41
Bounce Email Address | sys_blocked_email_address | (empty) | false | 2026-03-11 17:51:44
Bounce Email Address Log | sys_blocked_email_address_log | (empty) | false | 2026-03-11 17:51:44
Bounce Email Address Status | sys_bounce_email_address_status | (empty) | false | 2026-03-11 17:51:44
Boundary Connection Qualifier | cmdb_ci_qualifier_boundary_connection | Qualifier | false | 2026-03-11 18:06:22
Branding | sys_cs_branding_setup | Application File | false | 2026-03-11 19:04:08
Breakdown | pa_breakdowns | Application File | false | 2026-03-11 18:30:18
Breakdown | sn_bm_client_breakdown | (empty) | false | 2026-03-11 18:36:36

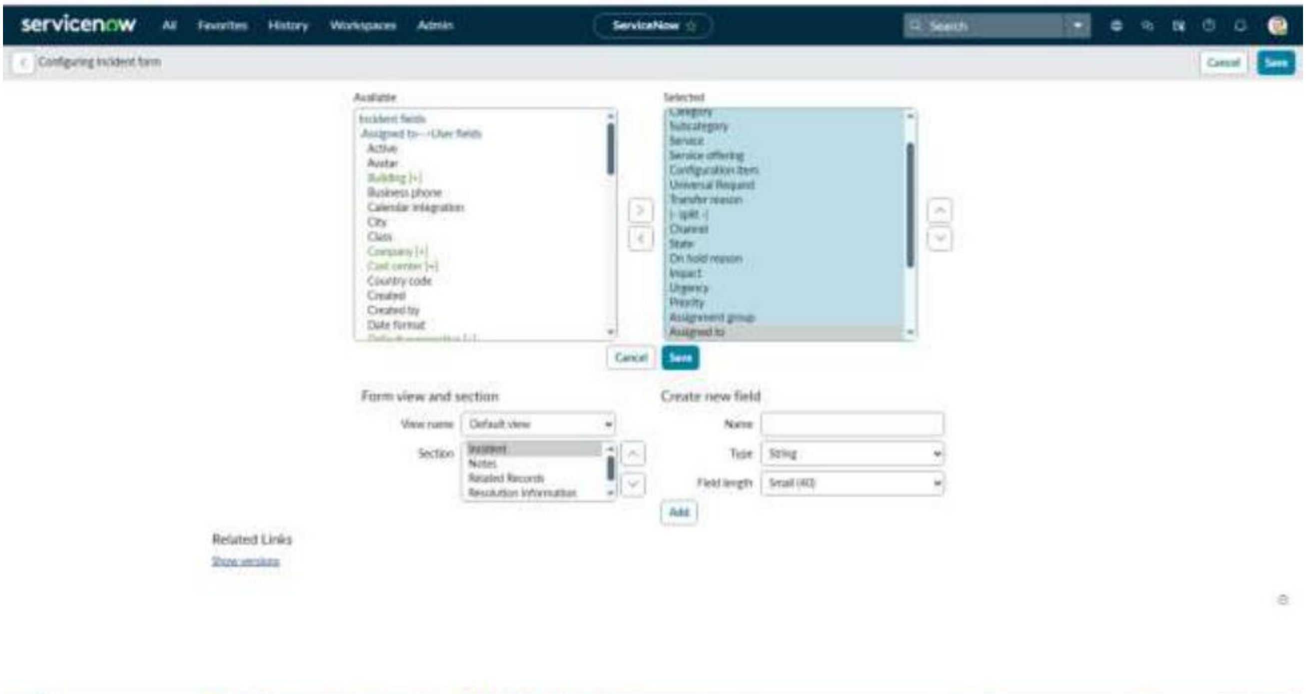

servicenow
All
Favorites
History
Workspaces
Admin
Table - Borrow Request
Search
Table
Borrow Request
Delete
Update
Delete All Records
Updated by | String | (empty) | 40 | false
Updates | Integer | (empty) | 40 | false
Updated | Date/Time | (empty) | 40 | false
Created by | String | (empty) | 40 | false
Book | Reference | Book [u_book] | 32 | false
Insert a new row...
Delete
Update
Delete All Records
Related Links
Form Builder
Design Form
Layout Form
Layout List
Show Form
Show List
Show Schema Map
Add to Service Catalog
Run Point Scan
Explore REST API


servicenow
All
Favorites
History
Workspaces
Admin
Table - Book
Search
Table
Book
Delete
Update
Delete All Records
A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. More Info
Label Book
Name u_book
Application Global
Remote Table
Columns
Controls
Application Access
Table Columns
for text
Search
1 to 6 of 6
New
Dictionary Entries
Column label
Type
Reference
Max length
Default value
Display
Updates Integer (empty) 40 false
Created by String (empty) 40 false
Updated by String (empty) 40 false
Sys ID Sys ID (GUID) (empty) 32 false


servicenow
All
Favorites
History
Workspaces
Admin
Report Sources
Search
Report Sources
Name
Search
Actions on selected rows...
New
Name
Table
Filter
Search
Search
Search
Change requests.Closed Change Request [change_request] Active = false
Change requests.Open Change Request [change_request] Active = true
Change requests.Open.Emergency Change Request [change_request] Active = true .and. Type = Emergency
Contracts.Active Contract [ast_contract] Active = true
Contracts.Active.Expiring Contract [ast_contract] State = Active .and. End date < javascri...
Feedback.Comments Knowledge Feedback [kb_feedback] Comments is not empty
Feedback.Flagged Knowledge Feedback [kb_feedback] Flagged = true
Feedback.Not Useful Knowledge Feedback [kb_feedback] Useful = No


servicenow
Transform Histories
Started
State
Completed
Run time
Set
Import set table
Total
Inserts
Updates
Ignored
Skipped
Errors
Transform Map
Complete
0 Seconds


servicenow
All
Favorites
History
Workspaces
Admin
Submit
Resolve
Number
Caller
Category
Subcategory
Service
Service offering
Configuration item
Channel
State
Impact
Urgency
Priority
Assignment group
Assigned to
Name
Department
Manager
Short description
Description
Related Search Results
Related Search
No results to display
Notes
Related Records
Resolution Information


servicenow
All
Favorites
History
Workspaces
Admin
ServiceNow
Cancel
Save
Available
Selected
Form view and section
View name
Default view
Section
Incident
Notes
Related Records
Resolution Information
Create new field
Name
Type
String
Field length
Add
Related Links
Show versions

# 8. ADVANTAGES & DISADVANTAGES

## 8.1. Advantages:

1. **Significant Time Savings (Reduced Processing Time):**
   By automating the book request and approval workflow, users and librarians no longer need to manually track requests. This reduces the time required to process and approve book requests efficiently.
2. **Elimination of Manual Effort:**
   The system automates request submission, tracking, and approval, eliminating the need for manual record-keeping and reducing human errors.
3. **Highly Maintainable (Low-Code/No-Code):**
   The workflow is built using standard platform features such as forms, workflows, and configurations. This makes the system easy to maintain without requiring complex coding.
4. **Upgrade Safe:**
   Since the solution uses native platform capabilities, it remains stable and compatible during system upgrades, ensuring uninterrupted functionality.
5. **Scalability:**
   The system can handle a large number of book requests efficiently, whether it is a small library or a large institution with thousands of users.

---

## 8.2. Disadvantages & Limitations:

1. **Dependence on Data Accuracy:**
   The system relies on correct input from users. If users enter incorrect book details or wrong information, it may lead to errors in request processing or approvals.
2. **Not Fully Real-Time in Some Cases:**
   If notifications or updates are processed in batches (e.g., scheduled jobs), there may be slight delays in reflecting request status or approvals.
3. **Possible UI Complexity:**
   Adding too many fields (like book details, user info, request status, history, etc.) can make the interface cluttered and difficult for users to navigate.
4. **Limited Direct Editing:**
   Some workflow data (like approved requests or system-generated details) may be read-only. Users or librarians may need to navigate to other sections to make corrections.
5. **Initial Setup Complexity:**
   Configuring workflows, roles, permissions, and request flows requires a good understanding of the platform. Incorrect setup may lead to workflow errors or access issues.

---

## 9. CONCLUSION:

The **Smart Library Request Workflow** project has successfully improved the efficiency and management of library operations by automating the book request and approval process. By replacing manual request handling with a structured digital workflow, the system ensures accurate tracking, faster processing, and reduced human errors.

The implementation of automated workflows enables users to easily submit book requests, while librarians can efficiently review, approve, or reject them. This streamlined process provides real-time visibility of request status, improving transparency and user satisfaction.

Additionally, the use of native platform features ensures that the system is highly scalable, maintainable, and secure. The low-code approach minimizes complexity and allows easy future enhancements without major changes.

Overall, this project demonstrates how automation can significantly enhance library management by reducing workload, improving response time, and creating a more organized and user-friendly system. It also provides a strong foundation for further improvements such as notifications, analytics, and integration with other systems.

## 10. FUTURE SCOPE:

1. **Real-Time Integration (Eliminating Manual Entry):**
   Future improvements can include integrating the system with external library databases or APIs. This will allow real-time updates of book availability, eliminating the need for manual data entry.
2. **Expansion Across Modules:**
   Currently, the workflow is focused on book requests. In future, it can be extended to other modules such as book returns, renewals, reservations, and fine management, providing a complete library management solution.
3. **Automated Allocation & Recommendations:**
   The system can be enhanced to automatically assign books based on availability and even recommend books to users based on their reading history or preferences.
4. **Smart Notifications & Alerts:**
   Future versions can include automated notifications such as request approval alerts, due date reminders, overdue warnings, and availability notifications for requested books.
5. **Advanced Validation & Error Handling:**
   To improve data accuracy, validation mechanisms can be added to verify user inputs and prevent incorrect or duplicate requests. Invalid entries can be flagged for review before processing.